NACA 4412 Calculations

- **>Defining the "NACA 4412" Geometry**
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- Using the NACA 4-digit convention, we define our shape parameters:
- **1st Digit (Max Camber, m):** 4% (0.04). This provides good lift.
- **2nd Digit (Max Camber Location, p): 40%** (0.4). The peak curvature is 0.4m from the front.
- **3rd/4th Digits (Thickness, t): 12%** (0.12). This is relatively thick, good for structural strength and fuel storage.
- **Chord Length (c): 1.0 meter** (for easy calculation).
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- **Step 1: Calculate the Thickness (yt)**

- *The "Flesh"*
- We need a shape that is rounded at the front (to catch air from different angles) and razor-sharp at the back (to reduce drag). To get this specific "teardrop" shape, we use a standard polynomial equation.
- Basically, for any point x, this formula tells you how much "meat" to add to the bone:
- $yt = 5t\ [\ 0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4\ ]$

- **Step 2: Calculate the Camber (yc)**

- *The "Spine"*
- Now we need to bend the wing. We don't use a single curve for this; we actually splice two different parabolas together.
- **The Front (0 to 40%):** We use one equation to curve the nose up to the peak.
- **The Back (40% to 100%):** We use a different equation to gently slope it back down to the tail.
- Since our peak (p) is at **0.4**, you just check where you are (x) and pick the right formula:
- **If you are in the Front (x<0.4):**
- $yc = (m / p^2)\ (\ 2px - x^2\ )$
- **If you are in the Back (x≥0.4):**
- $yc = (m / (1 - p)^2)\ [\ (1 - 2p) + 2px - x^2\ ]$

- **Step 3: Find the Slope (dyc /dx) & Angle (θ)**

- *The "Perpendicular Push"*
- Here is the tricky part where most people get stuck.
- You can't just add the thickness straight up and down. Since the spine is curved, "up" changes direction as you move along the wing. We need to add the thickness **perpendicular** to the spine.
- To do that, we first need the slope of our spine. Just like the camber, the slope formula changes depending on if you are in the front or back:
- **Front Section (0 ≤ x < p)**
- $dyc/dx = (2m / p^2)\ (\ p - x\ )$
- **Rear Section (p ≤ x ≤ 1)**
- $dyc/dx = (2m / (1 - p)^2)\ (\ p - x\ )$
- Once we have the slope, we turn it into an angle (θ) using arctan:
- $\theta = \arctan(dyc/dx)$
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- **Step 4: The Final Coordinates**

- *Putting it all together*
- Now we have everything we need. We take our point on the spine (x,yc) and push out by the thickness amount (yt) at our calculated angle (θ).
- **For the Upper Surface (Top of the wing):** We subtract a bit of x and add to y (tilting it back and up):
- $xU = x - yt \sin(\theta)$
- $yU = yc + yt \cos(\theta)$
- **For the Lower Surface (Bottom of the wing):** We add a bit of x and subtract from y (tilting it forward and down):
- $xL = x + yt \sin(\theta)$
- $yL = yc - yt \cos(\theta)$

NACA 4412

m(max camber)=	0.04									
p(max position)=	0.4									
t(thickness)=	0.12									
AOA(α)=	-10									
Conversion(Radians)=	-0.174532925199433									
	X	Yc	Chord X	Chord Y	Slope angle(θ)	Yt	xu	yu	xl	yl
	0	0	0	0	0.2	0	0	0	0	0
	0.0025	0.0004984375	1	0	0.19875	0.00871668416293359	0.000778942204259273	0.00904352631522348	0.00422105779574073	-0.00804665131522348
	0.005	0.00099375			0.1975	0.0122131474844947	0.00260355395775927	0.0129694761927888	0.00739644604224073	-0.0109819761927888
	0.0075	0.0014859375			0.19625	0.0148485818137621	0.00460463504713215	0.0160494962570378	0.0103953649528678	-0.0130776212570378
	0.01	0.001975			0.195	0.017037073971	0.00669878521074086	0.0186891817151933	0.0133012147892591	-0.0147391817151933
	0.0125	0.0024609375			0.19375	0.0189390266528367	0.00885347840081378	0.021045598131379	0.0161465215991862	-0.016123723131379
	0.015	0.00294375			0.1925	0.0206367117634073	0.011051922306764	0.0231992814651116	0.018948077693236	-0.0173117814651116
	0.0175	0.0034234375			0.19125	0.0221790089180859	0.0132840753127097	0.0251980652952031	0.0217159246872903	-0.0183511902952031
	0.02	0.0039			0.19	0.0235977712961143	0.0155433509891077	0.027073111352113	0.0244566490108923	-0.019273111352113
	0.0225	0.0043734375			0.18875	0.0249151289048086	0.0178251434076325	0.0288460622914691	0.0271748565923675	-0.0200991872914691
	0.025	0.00484375			0.1875	0.0261471986425573	0.0201260759719294	0.0305326748786699	0.0298739240280706	-0.0208451748786699
	0.0275	0.0053109375			0.18625	0.0273061512090666	0.0224435818165941	0.0321448438314854	0.0325564181834059	-0.0215229688314854
	0.03	0.005775			0.185	0.028401445417032	0.0247756525683511	0.0336918102706966	0.0352243474316489	-0.0221418102706966
	0.0325	0.0062359375			0.18375	0.0294406064581788	0.027120679517652	0.0351809235904017	0.037879320482348	-0.0227090485904017
	0.035	0.00669375			0.1825	0.030429738573433	0.0294773487202194	0.0366181432685262	0.0405226512797806	-0.0232306432685262
	0.0375	0.0071484375			0.18125	0.0313738751622281	0.0318445691747079	0.0380083815193175	0.0431554308252921	-0.0237115065193175
	0.04	0.0076			0.18	0.032277225216	0.0342214221382103	0.0393557444494633	0.0457785778617897	-0.0241557444494633
	0.0425	0.0080484375			0.17875	0.0331433512990601	0.036607124426485	0.0406637062066653	0.048392875573515	-0.0245668312066653
	0.045	0.00849375			0.1775	0.033975300974609	0.039001001244454	0.0419352375603079	0.050998998755546	-0.0249477375603079
	0.0475	0.0089359375			0.17625	0.0347757057471129	0.0414024656798265	0.0431729026900411	0.0535975343201735	-0.0253010276900411
	0.05	0.009375			0.175	0.0355468568261813	0.0438110029606874	0.0443789332913947	0.0561889970393126	-0.0256289332913947
	0.0525	0.0098109375			0.17375	0.0362907640219678	0.0462261581870565	0.0455552861750607	0.0587738418129435	-0.0259334111750607
	0.055	0.01024375			0.1725	0.0370092021471848	0.0486475266405174	0.046703688642135	0.0613524733594826	-0.026216188642135
	0.0575	0.0106734375			0.17125	0.03770374801983	0.0510747460372854	0.0478256746635606	0.0639252539627146	-0.0264787996635606
	0.06	0.0111			0.17	0.0383758102939395	0.0535074902671219	0.0489226140435179	0.0664925097328781	-0.0267226140435179
	0.0625	0.0115234375			0.16875	0.0390266537475586	0.0559454642828362	0.0499957361610399	0.0690545357171638	-0.0269488611610399
	0.065	0.01194375			0.1675	0.0396574192370084	0.0583883998911619	0.0510461494729308	0.0716116001088381	-0.0271586494729308
	0.0675	0.0123609375			0.16625	0.0402691402266691	0.0608360522572968	0.0520748576675893	0.0741639477427032	-0.0273529826675893
	0.07	0.012775			0.165	0.0408627565863046	0.0632881969800028	0.0530827731467807	0.0767118030199972	-0.0275327731467807
	0.0725	0.0131859375			0.16375	0.0414391261885181	0.0657446276269549	0.0540707283563738	0.0792553723730451	-0.0276988533563738
	0.075	0.01359375			0.1625	0.0419990347204226	0.0682051536444328	0.055039485371103	0.0817948463555672	-0.027851985371103
	0.0775	0.0139984375			0.16125	0.0425432040345356	0.0706695985738234	0.0559897440512543	0.0843304014261766	-0.0279928690512543
	0.08	0.0144			0.16	0.0430722992962286	0.0731377985213728	0.0569221490229577	0.0868622014786272	-0.0281221490229577
	0.0825	0.0147984375			0.15875	0.0435869351331478	0.0756096008383705	0.0578372956829798	0.0893903991616295	-0.0282404206829798
	0.085	0.01519375			0.1575	0.0440876809518246	0.0780848629772657	0.0587357353895833	0.0919151370227343	-0.0283482353895833
	0.0875	0.0155859375			0.15625	0.0445750655553144	0.080563451495729	0.0596179799703246	0.094436548504271	-0.0284461049703246
	0.09	0.015975			0.155	0.045049581171	0.0830452411857966	0.0604845056534981	0.0969547588142034	-0.0285345056534981
	0.0925	0.0163609375			0.15375	0.0455116869781123	0.0855301143093123	0.0613357565107801	0.0994698856906877	-0.0286138815107801
	0.095	0.01674375			0.1525	0.0459618122088822	0.0880179599241333	0.0621721474833313	0.101982040075867	-0.0286846474833313
	0.0975	0.0171234375			0.15125	0.0464003588846681	0.0905086732881898	0.0629940670513232	0.10449132671181	-0.0287471920513232
	0.1	0.0175			0.15	0.0468277042382395	0.0930021553306114	0.0638018795969142	0.106997844669389	-0.0288018795969142
	0.11	0.018975			0.145	0.0484318636263911	0.103001962349214	0.0668986150907204	0.116998037650786	-0.0289486150907204
	0.12	0.0204			0.14	0.049881816188064	0.113039336005006	0.0697937723095594	0.126960663994994	-0.0289937723095594
	0.13	0.021775			0.135	0.0511934370321155	0.123109859388312	0.0725026449040595	0.136890140611688	-0.0289526449040595
	0.14	0.0231			0.13	0.052379744863991	0.133209796683918	0.0750377590096364	0.146790203316082	-0.0288377590096364
	0.15	0.024375			0.125	0.0534516022043389	0.143335935746161	0.0774095550168151	0.156664064253839	-0.0286595550168151
	0.16	0.0256			0.12	0.054418208256	0.153485476172966	0.079626867104251	0.166514523827034	-0.028426867104251
	0.17	0.026775			0.115	0.055287454865753	0.163655947644441	0.0816972693016466	0.176344052355559	-0.0281472693016466
	0.18	0.0279			0.11	0.0560661893763429	0.17384514899585	0.083627330819811	0.18615485100415	-0.027827330819811
	0.19	0.028975			0.105	0.0567604125112336	0.184051101863138	0.0854228081009586	0.195948898136862	-0.0274728081009586
	0.2	0.03			0.1	0.0573754299023625	0.194272014801267	0.0870887917374681	0.205727985198733	-0.0270887917374681
	0.21	0.030975			0.095	0.0579159698809433	0.204506255078369	0.0886298205615661	0.215493744921631	-0.0266798205615661
	0.22	0.0319			0.09	0.0583862762814946	0.214752326194745	0.0900499714329479	0.225247673805255	-0.0262499714329479
	0.23	0.032775			0.085	0.0587901824472928	0.225008849738694	0.0913529307525883	0.234991150261306	-0.0258029307525883
	0.24	0.0336			0.08	0.0591311708918791	0.235274550574137	0.0925420520406969	0.244725449425863	-0.0253420520406969
	0.25	0.034375			0.075	0.059412421875	0.245548244620542	0.0936204027507151	0.254451755379458	-0.0248704027507151
	0.26	0.0351			0.07	0.0596368533111419	0.255828828673168	0.0945908026724877	0.264171171326832	-0.0243908026724877
	0.27	0.035775			0.065	0.0598071538290959	0.266115271846156	0.0954558556934958	0.273884728153844	-0.0239058556934958
	0.28	0.0364			0.06	0.0599258103666092	0.276406608318891	0.096217976264004	0.283593391681109	-0.023417976264004
	0.29	0.036975			0.055	0.059995131365294	0.286701931138302	0.0968794116015038	0.293298068861698	-0.0229294116015038
	0.3	0.0375			0.05	0.0600172663939703	0.297000386883732	0.0974422604391719	0.302999613116268	-0.0224422604391719
	0.31	0.037975			0.045	0.059994222850454	0.307301171041738	0.0979084889497014	0.312698828958262	-0.0219584889497014
	0.32	0.0384			0.04	0.0599278802564573	0.317603523969328	0.0982799443442184	0.322396476030672	-0.0214799443442184
	0.33	0.038775			0.035	0.0598200025564284	0.327906727348112	0.0985583665450178	0.332093272651888	-0.0210083665450178
	0.34	0.0391			0.03	0.0596722487507742	0.338210101050513	0.0987453982527143	0.341789898949487	-0.0205453982527143
	0.35	0.039375			0.025	0.0594861821311366	0.348513000353813	0.0988425936674	0.351486999646187	-0.0200925936674
	0.36	0.0396			0.02	0.059263278336	0.358814813449404	0.0988514260754161	0.361185186550596	-0.0196514260754161
	0.37	0.039775			0.015	0.0590049324057325	0.369114959203815	0.0987732944752995	0.370885040796185	-0.0192232944752995
	0.38	0.0399			0.01	0.0587124649848893	0.379412885135513	0.0986095293861035	0.380587114864487	-0.0188095293861035
	0.39	0.039975			0.005	0.0583871277944691	0.389708065577425	0.0983613979568922	0.390291934422575	-0.0184113979568922
	0.4	0.04			0	0.058030108476479	0.4	0.098030108476479	0.4	-0.018030108476479
	0.42	0.03995555555			-0.01	0.0572254789774376	0.420572245252242	0.0971781732828882	0.419427754747758	-0.017267062171777
	0.44	0.03982222222			-0.02	0.0563069482867822	0.441126063891306	0.0961179094947217	0.438873936108694	-0.0164734650502773
	0.46	0.0396			-0.03	0.0552820944953935	0.461658214076631	0.0948572194185853	0.458341785923369	-0.0156572194185853
	0.48	0.03928888888			-0.04	0.0541577863601279	0.482165733817563	0.0934033547964512	0.477834266182437	-0.0148255770186734
	0.5	0.03888888888			-0.05	0.0529402520005716	0.502645909815969	0.0917629793598349	0.497354090184031	-0.0139852015820571
	0.52	0.0384			-0.06	0.051635135338231	0.523096249589989	0.0899422199742495	0.516903750410011	-0.0131422199742495
	0.54	0.03782222222			-0.07	0.0502475482498186	0.543514456218631	0.0879467088523283	0.536485543781369	-0.0123022644078839
	0.56	0.03715555555			-0.08	0.04878208259198 2	0.563898405201517	0.0857816187202386	0.556101594798483	-0.0114705076091274
	0.58	0.0364			-0.09	0.0472428970038597	0.584246123042618	0.0834516923863982	0.575753876957382	-0.0106516923863982
	0.6	0.03555555555			-0.1	0.0456336906586779	0.60455576725266	0.080961267837949	0.59544423274734	-0.00985015672683785
	0.62	0.03462222222			-0.11	0.0439577538836464	0.624825607529965	0.0783142997473522	0.615174392470035	-0.00906985530290781
	0.64	0.0336			-0.12	0.042217982976	0.645054007929343	0.0755143780869043	0.634945992070657	-0.0083143780869043 1
	0.66	0.03248888888			-0.13	0.040416897596185	0.665239409867229	0.0725647444074388	0.654760590132771	-0.00758696662966097
	0.68	0.03128888888			-0.14	0.038556655805506	0.685380315841366	0.0694683062279658	0.674619684158634	-0.00689052845018798
	0.7	0.03			-0.15	0.0366390671267802	0.705475273767001	0.0662276498975175	0.694524726232999	-0.00622764989751747
	0.72	0.02862222222			-0.16	0.0346656039366859	0.725522861850393	0.0628450522232121	0.714477138149607	-0.00560060777876766
	0.74	0.02715555555			-0.17	0.0326374114430973	0.745521673935409	0.0593224911052321	0.734478326064591	-0.00501137999412088
	0.76	0.0256			-0.18	0.0305553164564671	0.765470305271092	0.0556616553768402	0.754529694728908	-0.00446165537684025
	0.78	0.02395555555			-0.19	0.0284198351287643	0.785367338657833	0.0518639540133367	0.774632661342167	-0.00395284290222552
	0.8	0.02222222222			-0.2	0.026231179804725	0.80521133093777	0.0479305248461773	0.79478866906223	-0.00348608040173292
	0.82	0.0204			-0.21	0.0239892651067799	0.825000799801541	0.043862242895944	0.814999200198459	-0.00306224289594404
	0.84	0.01848888888			-0.22	0.0216937133558867	0.84473421088888	0.039659728419408	0.83526578911112	-0.00268195064163022
	0.86	0.01648888888			-0.23	0.0193438594147605	0.86440996516499	0.0353233547507495	0.85559003483501	-0.00234557697297174
	0.88	0.0144			-0.24	0.0169387550269892	0.884026386558321	0.0308532560044303	0.875973613441679	-0.00205325600443032
	0.9	0.01222222222			-0.25	0.0144771727147185	0.903581709848433	0.0262493346967634	0.896418290151567	-0.00180489025231897
	0.92	0.00995555555			-0.26	0.0119576092885856	0.92307406879522	0.0215112693344928	0.91692593120478	-0.00160015822338174
	0.94	0.00760000000			-0.27	0.00937828901602896	0.942501484502928	0.0166385220113566	0.937498515497072	-0.00143852201135661
	0.96	0.00515555555			-0.28	0.00673716648775812	0.961861854014209	0.0116303460474206	0.958138145985791	-0.00131923493630946
	0.98	0.00262222222			-0.29	0.00403192921680022	0.981152939131009	0.00648579370072175	0.978847060868991	-0.00124134925627729
	1	0				0.00126	1	0.00126	1	-0.00126